Smart Factory Iot Dashboard With Equipment Control

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Abstract:

The Smart Factory IoT Dashboard with Equipment Control is a real-time monitoring and automation system based on the ESP32 microcontroller and Firebase Realtime Database. It collects data from sensors such as temperature, humidity, and IR sensors and displays it on a web-based dashboard. The system also enables remote control of industrial equipment like fans, lights, motors, and buzzers through the dashboard. This solution improves operational efficiency, enhances safety, and supports smart manufacturing using IoT technologies.

Introduction:

In recent years, the adoption of Internet of Things (IoT) technology has significantly transformed traditional industrial systems into smart and automated environments.

This project presents a **Smart Factory IoT Dashboard with Equipment Control** using the ESP32 microcontroller and Firebase Realtime Database. The system collects real-time data from sensors such as temperature, humidity, and infrared sensors, and transmits it over Wi-Fi to a cloud-based platform. The data is then visualized on a web dashboard developed using HTML, CSS, and JavaScript.

In addition to monitoring, the system enables users to remotely control industrial equipment such as fans, lights, motors, and buzzers through the dashboard. Control commands are sent via Firebase and executed by the ESP32 using relay modules. This integrated approach enhances operational efficiency, reduces manual intervention, and improves overall system reliability.

The proposed system demonstrates a cost-effective and scalable solution for implementing smart factory automation using IoT technologies.

Objectives:

The objective of this project is to design and develop a Smart Factory IoT system using the ESP32 microcontroller for real-time monitoring and control of industrial equipment. The system collects sensor data such as temperature and humidity, stores it in Firebase Realtime Database, and displays it on a web-based dashboard. It also enables remote control of devices like fans, lights, conveyors, and buzzers, aiming to improve efficiency, safety, and automation in industrial environments.

Proposed Work: (Font 16)

The proposed system aims to develop a Smart Factory IoT Dashboard using the ESP32 microcontroller integrated with sensors and relay modules for real-time monitoring and control of industrial equipment. Sensor data such as temperature and humidity is collected and transmitted to Firebase Realtime Database via Wi-Fi, where it is displayed on a web-based dashboard. The system also allows users to remotely control devices like fans, lights, conveyors, and buzzers through the dashboard. This approach provides a cost-effective and scalable solution for improving automation, monitoring, and efficiency in industrial environments.

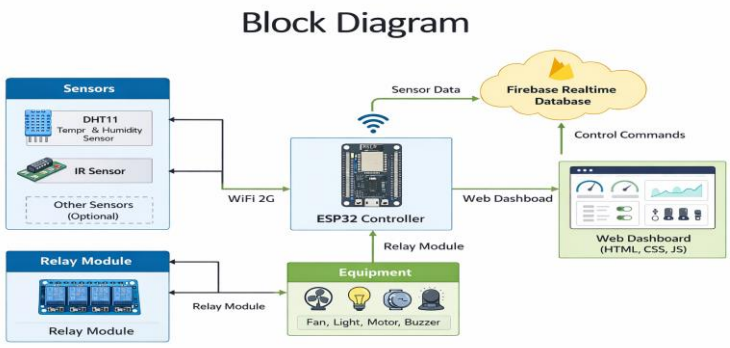


Figure – Block Diagram

Feature:

- Firebase-based cloud data storage
- Web dashboard for live data visualization
- Remote control of equipment (fan, light, conveyor, buzzer)
- Wi-Fi enabled communication using ESP32
- Low-cost and scalable IoT solution

Results:

- Successful real-time monitoring of temperature and humidity data
- Reliable remote control of equipment through the web dashboard
- Stable data communication using Firebase Realtime Database
- Improved system efficiency with reduced manual intervention

Applications:

- Smart factory automation systems
- Industrial monitoring and control systems
- Remote equipment management in manufacturing units
- IoT-based smart home and automation systems

Conclusion:

- Real-time monitoring and control achieved using ESP32 and Firebase
- Easy remote operation through web dashboard
- Improves efficiency and reduces manual work
- Supports smart and scalable industrial automation